

Function Code Documentation (All Apps)

This document lists function/method definitions found in the repository apps, along with location, signature, and code snippet.

Apps Covered

- **Employee**: 9 functions/methods
- **Department**: 5 functions/methods
- **Role**: 2 functions/methods
- **Project**: 2 functions/methods
- **Leave**: 5 functions/methods
- **Attendance**: 7 functions/methods
- **Dashboard**: 7 functions/methods
- **Login_App**: 2 functions/methods
- **E_Managements**: 1 functions/methods

Employee App

File: 'Employee/models.py'

1. 'BusinessProfile.__str__' (method)

```
- Signature:** ' __str__(self)'  
- Location:** 'Employee/models.py:13-14'  
- Code:**  
"""python  
def __str__(self):  
    return self.business_name  
"""
```

2. 'Employee.__str__' (method)

```
- Signature:** ' __str__(self)'  
- Location:** 'Employee/models.py:66-67'  
- Code:**  
"""python  
def __str__(self):  
    return self.username  
"""
```

File: 'Employee/servies.py'

1. 'Get_Employee_view' (function)

```
- Signature:** 'Get_Employee_view(request)'  
- Location:** 'Employee/servies.py:7-9'  
- Code:**  
"""python  
def Get_Employee_view(request):  
    get_Employee = Employee.objects.filter(roal='employee', is_active=True)  
    return get_Employee  
"""
```

File: 'Employee/views.py'

1. 'Create_Employeeview' (function)

```
- Signature:** 'Create_Employeeview(request)'  
- Location:** 'Employee/views.py:15-79'  
- Code:**  
"""python  
def Create_Employeeview(request):  
    business = request.user.business_profile
```

```

if request.method == 'POST':
    first_name = request.POST.get('first_name')
    last_name = request.POST.get('last_name')
    email = request.POST.get('email')
    phone = request.POST.get('phone')
    hire_date = request.POST.get('hire_date')

    salary = request.POST.get('salary')
    salary = Decimal(salary) if salary else None

    job_id = request.POST.get('job')
    department_id = request.POST.get('department')

    department = Department.objects.filter(
        id=department_id,
        business_profile=business
    ).first()

    job = Job.objects.filter(
        id=job_id,
        business_profile=business
    ).first()

    if Employee.objects.filter(email=email).exists():
        messages.error(request, 'An employee with this email already exists.')
        return redirect('Employee:create_employee')
    try:
        employee = Employee.objects.create_user(
            username = email,
            email=email,
            password='Test@1234',
            first_name=first_name,
            last_name=last_name,
        )
    except IntegrityError :
        messages.error(request, 'An employee with this username already exists.')
        return redirect('Employee:create_employee')
    employee.phone = phone
    employee.hire_date = hire_date
    employee.salary = salary
    employee.business_profile = business
    employee.role = 'employee'
    employee.department = department
    employee.job = job
    employee.save()
    messages.success(request, 'Employee created successfully.')
    return redirect('Dashboard:dashboard_view')

# ? Correct filtering (NEW model design)
departments = Department.objects.filter(
    business_profile=business
)

jobs = Job.objects.filter(
    business_profile=business
)

return render(request, 'Create_Employee.html', {
    'departments': departments,
    'jobs': jobs
})

```

```
##### 2. 'Employee_List_view' (function)
- **Signature:** 'Employee_List_view(request)'
- **Location:** 'Employee/views.py:82-87'
- **Code:**
'''python
def Employee_List_view(request):
    Employees = Get_Employee_view(request)
    context = {
        'Employees': Employees
    }
    return render(request, 'Employee_Dashboard.html', context)
'''
```

```
##### 3. 'Owner_register' (function)
- **Signature:** 'Owner_register(request)'
- **Location:** 'Employee/views.py:93-113'
- **Code:**
'''python
def Owner_register(request):
    if request.method == 'POST':
        owner = Employee.objects.create_user(
            first_name=request.POST.get('first_name'),
            last_name=request.POST.get('last_name'),
            email=request.POST.get('email'),
            phone=request.POST.get('phone'),
            role='owner',
            username=request.POST.get('username'),
            password=request.POST.get('password'),
        )
        business = BusinessProfile.objects.create(
            business_name=request.POST.get('business_name'),
            business_address=request.POST.get('business_address'),
            business_phone=request.POST.get('business_phone'),
            business_email=request.POST.get('business_email'),
        )
        owner.business_profile = business
        owner.save()
        return redirect('Home')
    return render(request, 'Owner_register.html')
'''
```

```
##### 4. 'Employee_Detail_view' (function)
- **Signature:** 'Employee_Detail_view(request, employee_id)'
- **Location:** 'Employee/views.py:118-123'
- **Code:**
'''python
def Employee_Detail_view(request, employee_id):
    employee = Employee.objects.get(id=employee_id)
    context = {
        'employee': employee
    }
    return render(request, 'Employee_Detail.html', context)
'''
```

```
##### 5. 'Employee_Delete_view' (function)
- **Signature:** 'Employee_Delete_view(request, employee_id)'
- **Location:** 'Employee/views.py:125-129'
- **Code:**
'''python
def Employee_Delete_view(request, employee_id):
    employee = Employee.objects.get(id=employee_id)
    employee.delete()
'''
```

```

    messages.success(request, 'Employee deleted successfully.')
    return redirect('Dashboard:dashboard_view')
"""

##### 6. 'Employee_Update_view' (function)
- **Signature:** 'Employee_Update_view(request, employee_id)'
- **Location:** 'Employee/views.py:132-182'
- **Code:**
"""python
def Employee_Update_view(request, employee_id):
    employee = Employee.objects.get(id=employee_id)
    business = request.user.business_profile

    if request.method == 'POST':
        form = EmployeeUpdateForm(request.POST, instance=employee)
        if form.is_valid():
            employee = form.save(commit=False)

            job_id = request.POST.get('job')
            department_id = request.POST.get('department')

            # Filter department and job by business profile for security
            if department_id:
                employee.department = Department.objects.filter(
                    id=department_id,
                    business_profile=business
                ).first()

            if job_id:
                employee.job = Job.objects.filter(
                    id=job_id,
                    business_profile=business
                ).first()

            employee.save()
            messages.success(request, 'Employee updated successfully.')
            return redirect('Employee:employee_detail', employee_id=employee.id)
        else:
            messages.error(request, 'Please correct the errors below.')
    else:
        form = EmployeeUpdateForm(instance=employee)

    # Filter departments and jobs by business profile
    departments = Department.objects.filter(
        business_profile=business
    )

    jobs = Job.objects.filter(
        business_profile=business
    )

    # Update form querysets to show only business-specific options
    form.fields['department'].queryset = departments
    form.fields['job'].queryset = jobs

    context = {
        'form': form,
        'employee': employee,
    }
    return render(request, 'Employee_Update.html', context)
"""

```

Department App

File: 'Department/models.py'

1. 'Department.total_employees' (method)

- **Signature:** 'total_employees(self)'
- **Location:** 'Department/models.py:13-14'
- **Code:**

```
"""python
def total_employees(self):
    return self.employees.count()
"""
```

2. 'Department.__str__' (method)

- **Signature:** '__str__(self)'
- **Location:** 'Department/models.py:16-17'
- **Code:**

```
"""python
def __str__(self):
    return self.name
"""
```

File: 'Department/views.py'

1. 'Department_view' (function)

- **Signature:** 'Department_view(request)'
- **Location:** 'Department/views.py:7-18'
- **Code:**

```
"""python
def Department_view(request):
    if request.method == 'POST':
        form = DepartmentForm(request.POST)
        if form.is_valid():
            depart = form.save(commit=False)
            depart.business_profile = request.user.business_profile
            depart.save()
            return redirect('Dashboard:dashboard_view')
    context = {
        'form': DepartmentForm()
    }
    return render(request, 'Department.html', context)
"""
```

2. 'Department_detail_view' (function)

- **Signature:** 'Department_detail_view(request, department_id)'
- **Location:** 'Department/views.py:23-28'
- **Code:**

```
"""python
def Department_detail_view(request, department_id):
    department = Department.objects.get(id=department_id)
    context = {
        'department': department
    }
    return render(request, 'Department_Detail.html', context)
"""
```

3. 'delete_department' (function)

- **Signature:** 'delete_department(request, department_id)'
- **Location:** 'Department/views.py:34-38'
- **Code:**

```
"""python
def delete_department(request, department_id):
```

```

    department = Department.objects.get(id=department_id)
    department.delete()
    messages.success(request, 'Department deleted successfully.')
    return redirect('Dashboard:dashboard_view')
"""

```

Role App

File: 'Role/models.py'

```

##### 1. 'Job.__str__' (method)
- **Signature:** ' __str__(self)'
- **Location:** 'Role/models.py:10-11'
- **Code:**
"""python
def __str__(self):
    return self.title
"""

```

File: 'Role/views.py'

```

##### 1. 'Job_view' (function)
- **Signature:** 'Job_view(request)'
- **Location:** 'Role/views.py:6-19'
- **Code:**
"""python
def Job_view(request):
    if request.method == 'POST':
        form = JobForm(request.POST)
        if form.is_valid():
            job = form.save(commit=False)
            job.business_profile = request.user.business_profile
            job.save()
            return redirect('Dashboard:dashboard_view')
    else:
        form = JobForm()
        context = {
            'form': form
        }
    return render(request, 'Job.html', context)
"""

```

Project App

File: 'Project/models.py'

```

##### 1. 'Project.__str__' (method)
- **Signature:** ' __str__(self)'
- **Location:** 'Project/models.py:22-23'
- **Code:**
"""python
def __str__(self):
    return self.name
"""

```

File: 'Project/views.py'

```

##### 1. 'Project_view' (function)
- **Signature:** 'Project_view(request)'

```

```

- **Location:** 'Project/views.py:7-13'
- **Code:**
"""python
def Project_view(request):
    if request.method == 'POST':
        form = ProjectForm(request.POST)
        if form.is_valid():
            form.save()
            return redirect('Home')
    return render(request, 'Project.html', {'form': form})
"""

---

## Leave App

### File: 'Leave/models.py'

#### 1. 'Leave.clean' (method)
- **Signature:** 'clean(self)'
- **Location:** 'Leave/models.py:55-62'
- **Code:**
"""python
def clean(self):
    if self.end_date < self.start_date:
        raise ValidationError("End date cannot be before start date")

    if self.employee.business_profile != self.business_profile:
        raise ValidationError(
            "Employee must belong to the same business as the leave"
        )
"""

#### 2. 'Leave.__str__' (method)
- **Signature:** '.__str__(self)'
- **Location:** 'Leave/models.py:64-65'
- **Code:**
"""python
def __str__(self):
    return f'{self.employee.username} ({self.leave_type}) - {self.start_date} to {self.end_date}'
"""

### File: 'Leave/views.py'

#### 1. 'Leave_view' (function)
- **Signature:** 'Leave_view(request)'
- **Location:** 'Leave/views.py:8-16'
- **Code:**
"""python
def Leave_view(request):
    if request.method == 'POST':
        form = LeaveForm(request.POST)
        if form.is_valid():
            form.save()
            return redirect('Home')
    else:
        form = LeaveForm()
    return render(request, 'Leave.html', {'form': form})
"""

#### 2. 'manage_leave' (function)
- **Signature:** 'manage_leave(request, leave_id, action)'

```

```

- **Location:** 'Leave/views.py:21-31'
- **Code:**
'''python
def manage_leave(request, leave_id, action):
    leave = Leave.objects.get(pk=leave_id)

    if action == 'approve':
        leave.status = 'approved'
        leave.approved_by = request.user
    elif action == 'reject':
        leave.status = 'rejected'

    leave.save()
    return redirect('Dashboard:dashboard_view')
'''

#### 3. 'Employee_leave_apply_view' (function)
- **Signature:** 'Employee_leave_apply_view(request)'
- **Location:** 'Leave/views.py:37-59'
- **Code:**
'''python
def Employee_leave_apply_view(request):
    if request.method == 'POST':
        leave_type = request.POST.get('leave_type')
        start_date = request.POST.get('start_date')
        end_date = request.POST.get('end_date')
        reason = request.POST.get('reason')

        # ? yahan sirf Employee object lena hai
        employee = request.user.employee

        Leave.objects.create(
            employee=employee,          # ? correct
            business_profile=employee.business_profile,
            leave_type=leave_type,
            start_date=start_date,
            end_date=end_date,
            reason=reason,
            status='pending'
        )
        employee.save()
        return HttpResponse("Leave request submitted successfully")

    return HttpResponse("Invalid request method")
'''

---

## Attendance App

#### File: 'Attendance/forms.py'

#### 1. 'AttendanceForm.save' (method)
- **Signature:** 'save(self, commit)'
- **Location:** 'Attendance/forms.py:11-21'
- **Code:**
'''python
def save(self, commit=True):
    attendance = super().save(commit=False)

    # safety (model already handles this, but extra protection)
    if not attendance.business_profile and attendance.employee:
        attendance.business_profile = attendance.employee.business_profile
'''

```



```

        if commit:
            attendance.save()

    return attendance
"""

```

File: 'Attendance/models.py'

1. 'Attendance.clean' (method)

```

- **Signature:** 'clean(self)'
- **Location:** 'Attendance/models.py:40-49'
- **Code:**
"""python
def clean(self):
    if self.check_in and self.check_out:
        if self.check_out < self.check_in:
            raise ValidationError("Check-out time cannot be before check-in")

    if self.employee_id and self.business_profile_id:
        if self.employee.business_profile_id != self.business_profile_id:
            raise ValidationError(
                "Employee must belong to the same business as attendance record"
            )
"""

```

2. 'Attendance.save' (method)

```

- **Signature:** 'save(self, *args, **kwargs)'
- **Location:** 'Attendance/models.py:51-54'
- **Code:**
"""python
def save(self, *args, **kwargs):
    if not self.business_profile and self.employee:
        self.business_profile = self.employee.business_profile
    super().save(*args, **kwargs)
"""

```

3. 'Attendance.Total_hours' (method)

```

- **Signature:** 'Total_hours(self)'
- **Location:** 'Attendance/models.py:56-60'
- **Code:**
"""python
def Total_hours(self):
    if self.check_in and self.check_out:
        delta = datetime.combine(date.min, self.check_out) - datetime.combine(date.min,
self.check_in)
        return delta
    return None
"""

```

4. 'Attendance.__str__' (method)

```

- **Signature:** '__str__(self)'
- **Location:** 'Attendance/models.py:61-62'
- **Code:**
"""python
def __str__(self):
    return f'{self.employee.username} - {self.date}'
"""

```

File: 'Attendance/views.py'

1. 'attendance_create' (function)

```

- **Signature:** 'attendance_create(request)'

```

```

- **Location:** 'Attendance/views.py:7-21'
- **Code:**
'''python
def attendance_create(request):
    if request.method == 'POST':
        form = AttendanceForm(request.POST)
        if form.is_valid():
            attendance = form.save(commit=False)

            # ? Assign business from logged-in user
            attendance.business_profile = request.user.business_profile
            attendance.save()

            return redirect('Dashboard:dashboard_view') # change as needed
    else:
        form = AttendanceForm()

    return render(request, 'Attendance.html', {'form': form})
'''

```

```

##### 2. 'Attendance_list_view' (function)
- **Signature:** 'Attendance_list_view(request)'
- **Location:** 'Attendance/views.py:26-31'
- **Code:**
'''python
def Attendance_list_view(request):
    attendance_records = Attendance.objects.all()
    context = {
        'attendance_records': attendance_records
    }
    return context
'''

```

Dashboard App

File: 'Dashboard/servies.py'

```

##### 1. 'count_employee_view' (function)
- **Signature:** 'count_employee_view(request)'
- **Location:** 'Dashboard/servies.py:13-15'
- **Code:**
'''python
def count_employee_view(request):
    employee_count = Employee.objects.filter(business_profile=request.user.business_profile,
role='employee', is_active=True).count()
    return employee_count
'''

```

```

##### 2. 'get_recent_added_employees' (function)
- **Signature:** 'get_recent_added_employees(user)'
- **Location:** 'Dashboard/servies.py:17-21'
- **Code:**
'''python
def get_recent_added_employees(user):
    return Employee.objects.filter(
        business_profile=user.business_profile,
        role='employee'
    ).order_by('-date_joined')[:3]
'''

```

```

##### 3. 'fiend_Total_deparment' (function)

```

```
- **Signature:** 'fiend_Total_deparment(request)'
- **Location:** 'Dashboard/servies.py:29-31'
- **Code:**
'''python
def fiend_Total_deparment(request):
    total_departments =
    Department.objects.filter(business_profile=request.user.business_profile)
    return total_departments
'''
```

```
##### 4. 'today_attendance_view' (function)
- **Signature:** 'today_attendance_view(request)'
- **Location:** 'Dashboard/servies.py:42-45'
- **Code:**
'''python
def today_attendance_view(request):
    today = timezone.now().date()
    today_attendance =
    Attendance.objects.filter(business_profile=request.user.business_profile, date=today)
    return today_attendance
'''
```

```
##### 5. 'Get_pending_leave_view' (function)
- **Signature:** 'Get_pending_leave_view(request)'
- **Location:** 'Dashboard/servies.py:53-55'
- **Code:**
'''python
def Get_pending_leave_view(request):
    pending_leaves = Leave.objects.filter(status='pending',
    business_profile=request.user.business_profile)
    return pending_leaves
'''
```

File: 'Dashboard/views.py'

```
##### 1. 'Dashboard_view' (function)
- **Signature:** 'Dashboard_view(request)'
- **Location:** 'Dashboard/views.py:7-19'
- **Code:**
'''python
def Dashboard_view(request):
    count = count_employee_view(request)
    departments = fiend_Total_deparment(request)
    context = {
        'total_employees': count,
        'Total_departments': departments,
        'today_attendance': today_attendance_view(request),
        'pending_leaves': Get_pending_leave_view(request).count(),
        'pending_leave_list': Get_pending_leave_view(request),
        'recentadd_employees':get_recent_added_employees(request.user),

    }
    return render(request, 'dashboard.html', context)
'''
```

```
##### 2. 'Employee_Dashboard_view' (function)
- **Signature:** 'Employee_Dashboard_view(request)'
- **Location:** 'Dashboard/views.py:22-26'
- **Code:**
'''python
def Employee_Dashboard_view(request):
    context = {
        'leave_types': Leave.objects.values_list('leave_type', flat=True).distinct()
    }
    return render(request, 'employee_dashboard.html', context)
'''
```

```
    }
    return render(request, 'Employee_Dashboard.html', context)

```

Login_App App

File: 'Login_App/views.py'

1. 'login_view' (function)

```
- **Signature:** 'login_view(request)'
- **Location:** 'Login_App/views.py:7-20'
- **Code:**
```

```
"""python
```

```
def login_view(request):
    if request.method == 'POST':
        username = request.POST.get('username')
        password = request.POST.get('password')
        user = authenticate(request, username=username, password=password)
        if user is None:
            return HttpResponse("Invalid username or password")
        if user.role == 'owner':
            login(request, user)
            return redirect('Dashboard:dashboard_view')
        elif user.role == 'employee':
            login(request, user)
            return redirect('Dashboard:employee_dashboard')
    return render(request, 'Login.html')
"""
```

2. 'logout_view' (function)

```
- **Signature:** 'logout_view(request)'
- **Location:** 'Login_App/views.py:22-24'
- **Code:**
```

```
"""python
```

```
def logout_view(request):
    logout(request)
    return redirect('Home')
```

```
"""
```

E_Managements App

File: 'E_Managements/views.py'

1. 'Home_view' (function)

```
- **Signature:** 'Home_view(request)'
- **Location:** 'E_Managements/views.py:5-6'
- **Code:**
```

```
"""python
```

```
def Home_view(request):
    return render(request, "Home.html")
```

```
"""
```